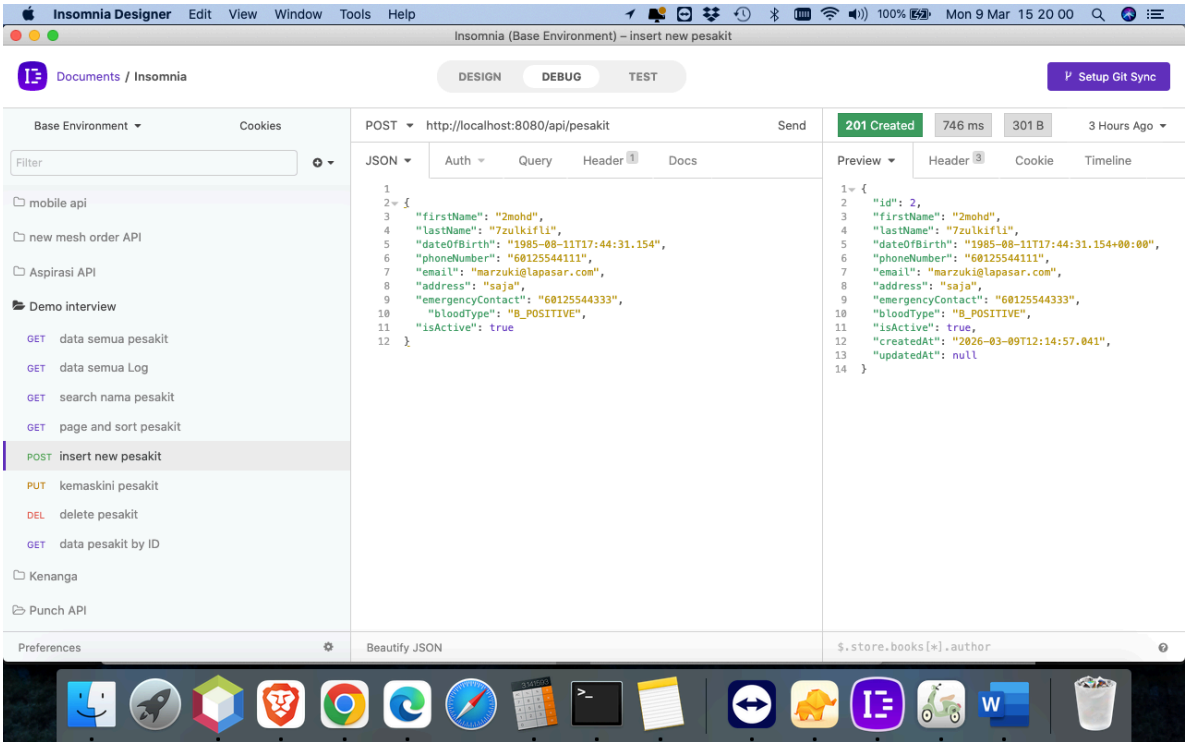
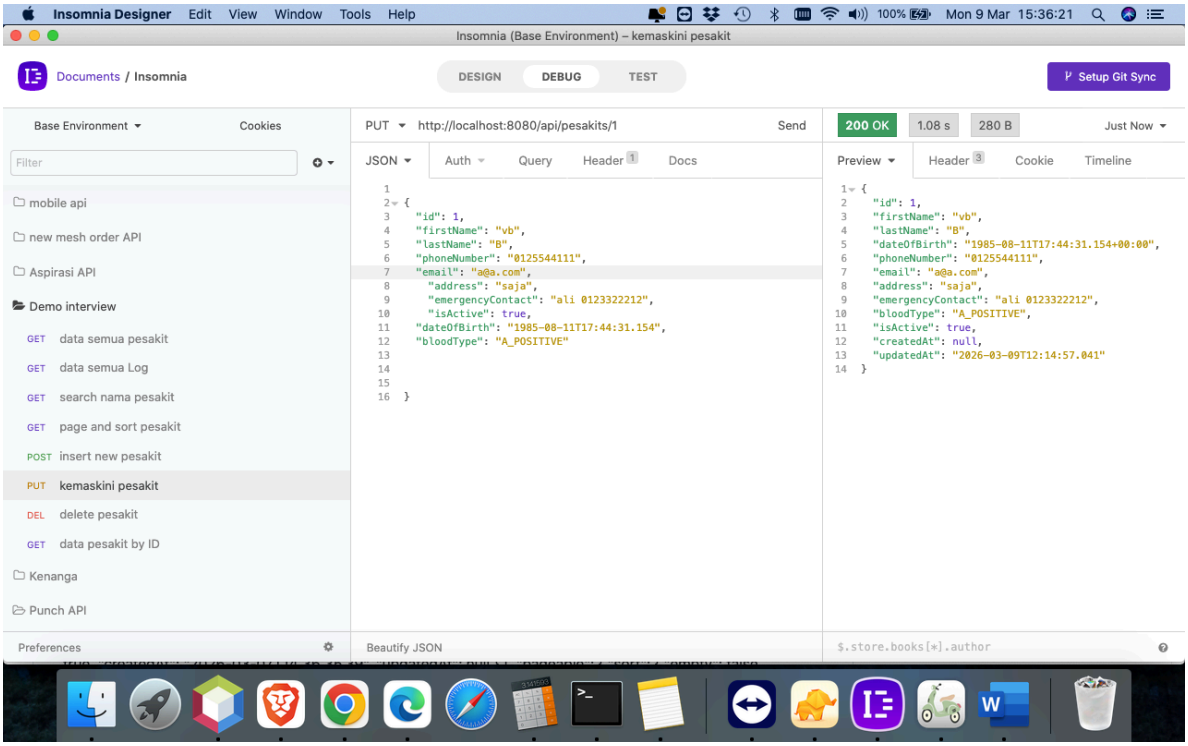
Mini hospital management system API spring boot 2.5

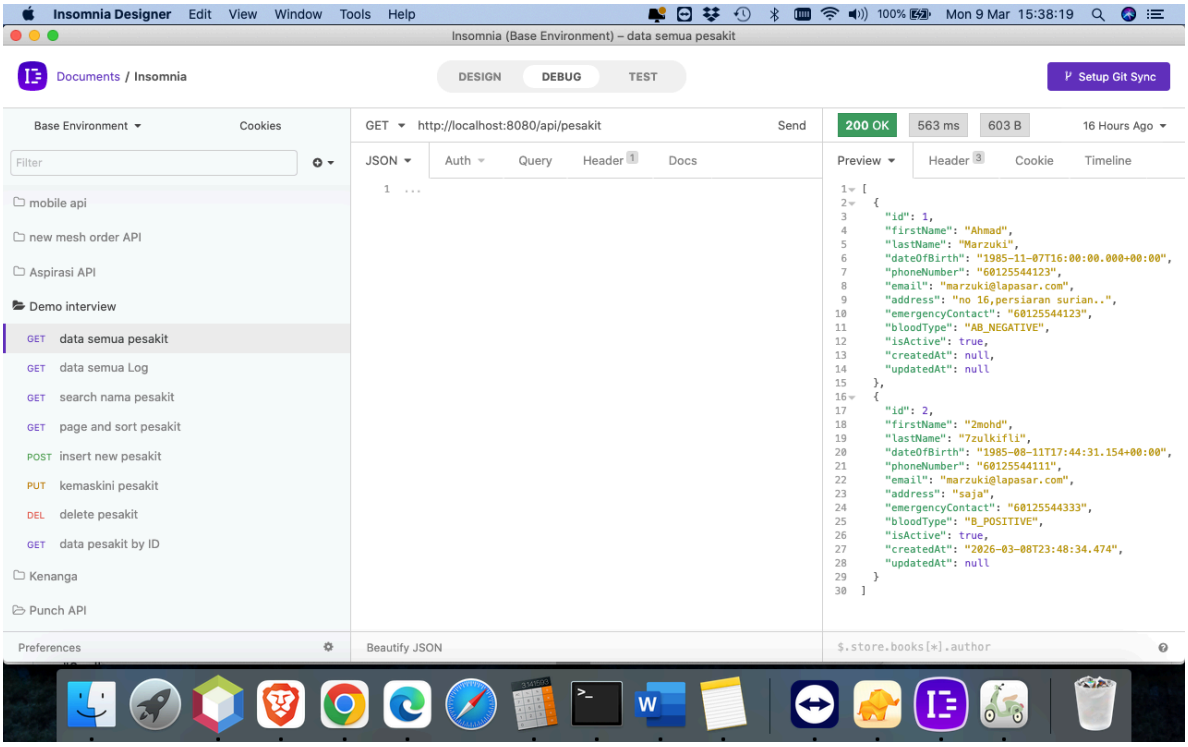
https://github.com/marzuki234/hospital-patient--management-api
README.md ada dalam git file tu



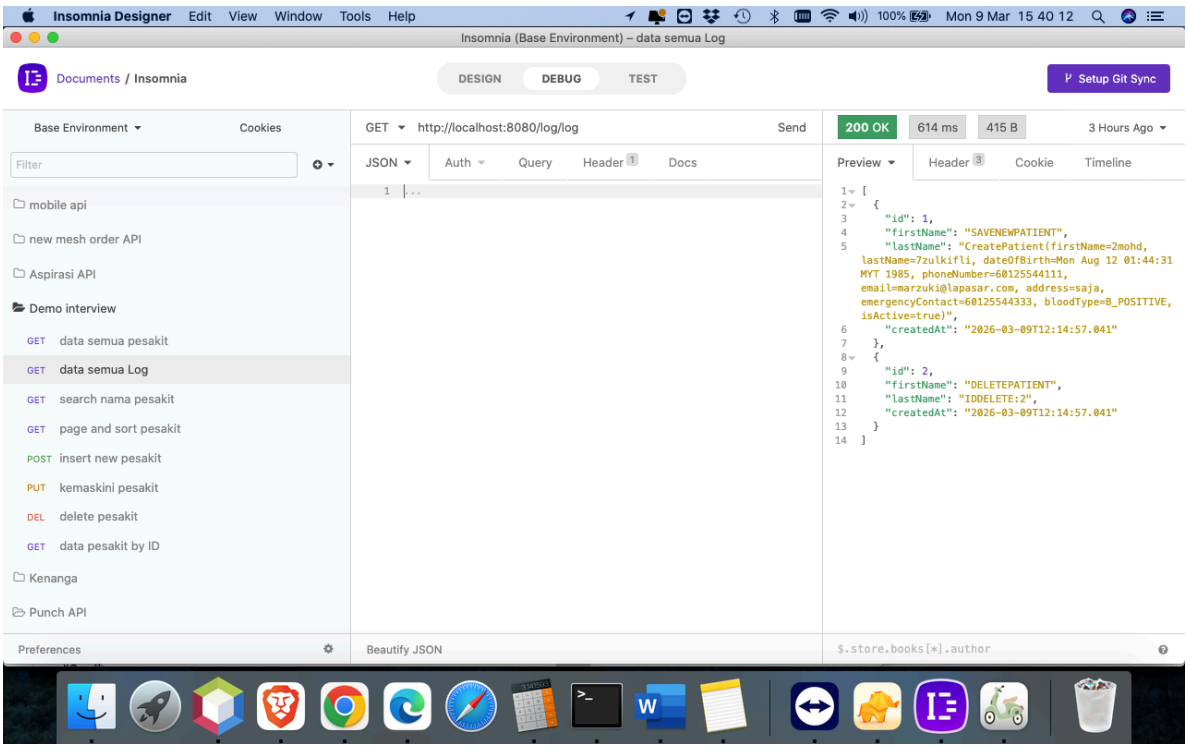
insert patient



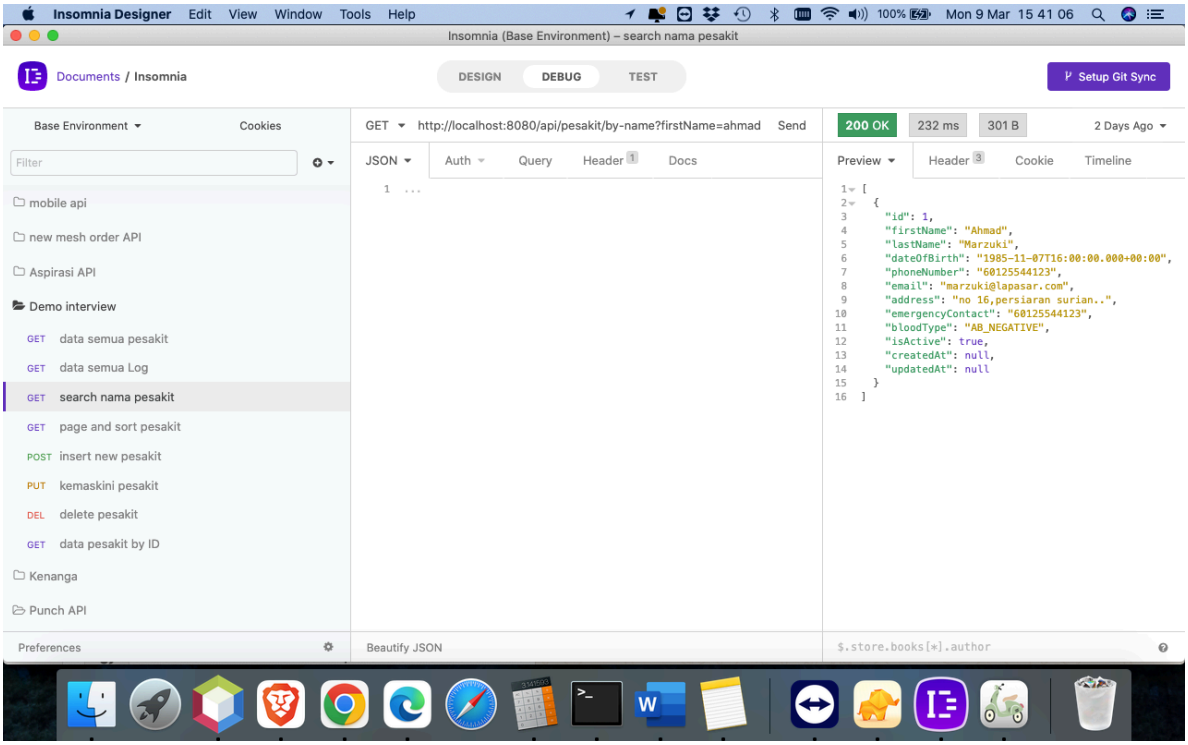
Kemaskini patient



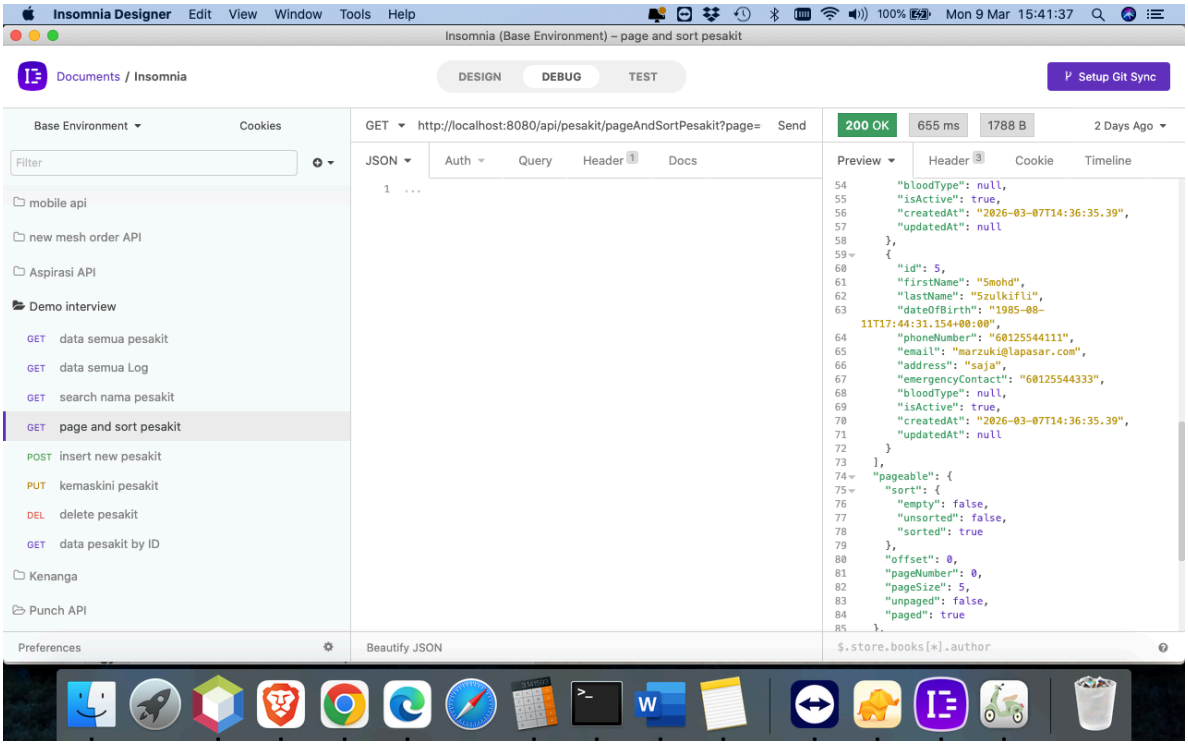
All patients list



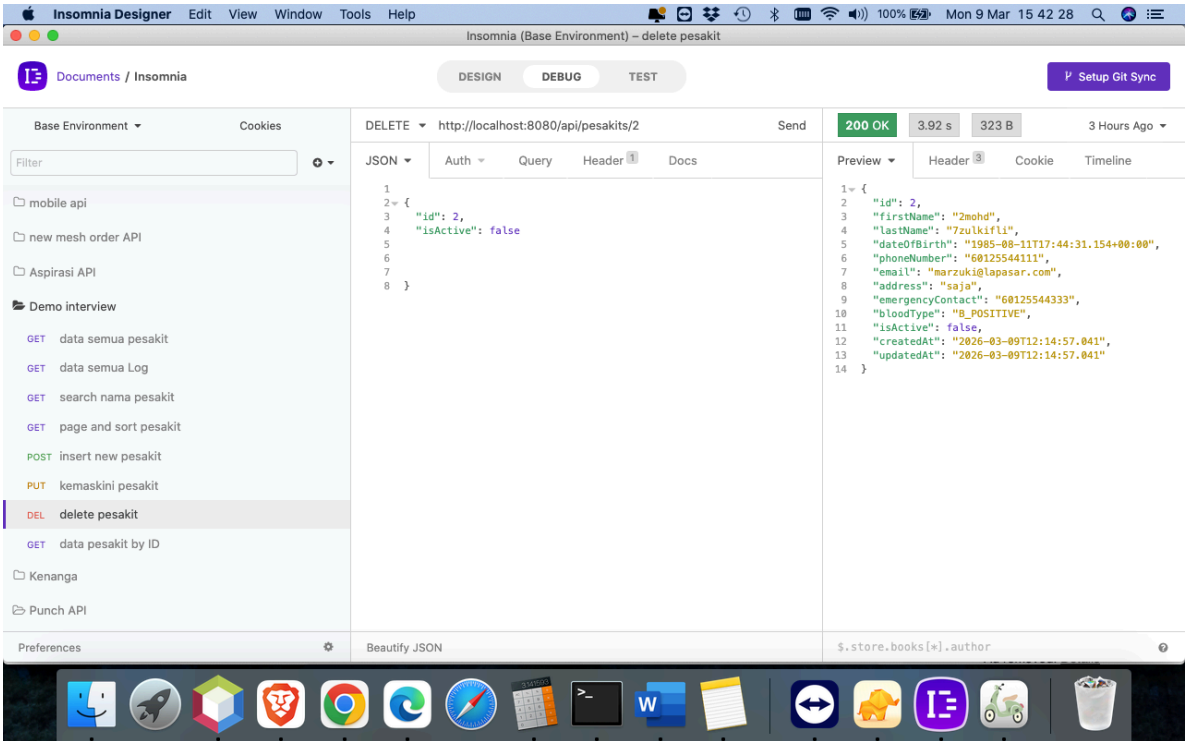
Audit trail log lists



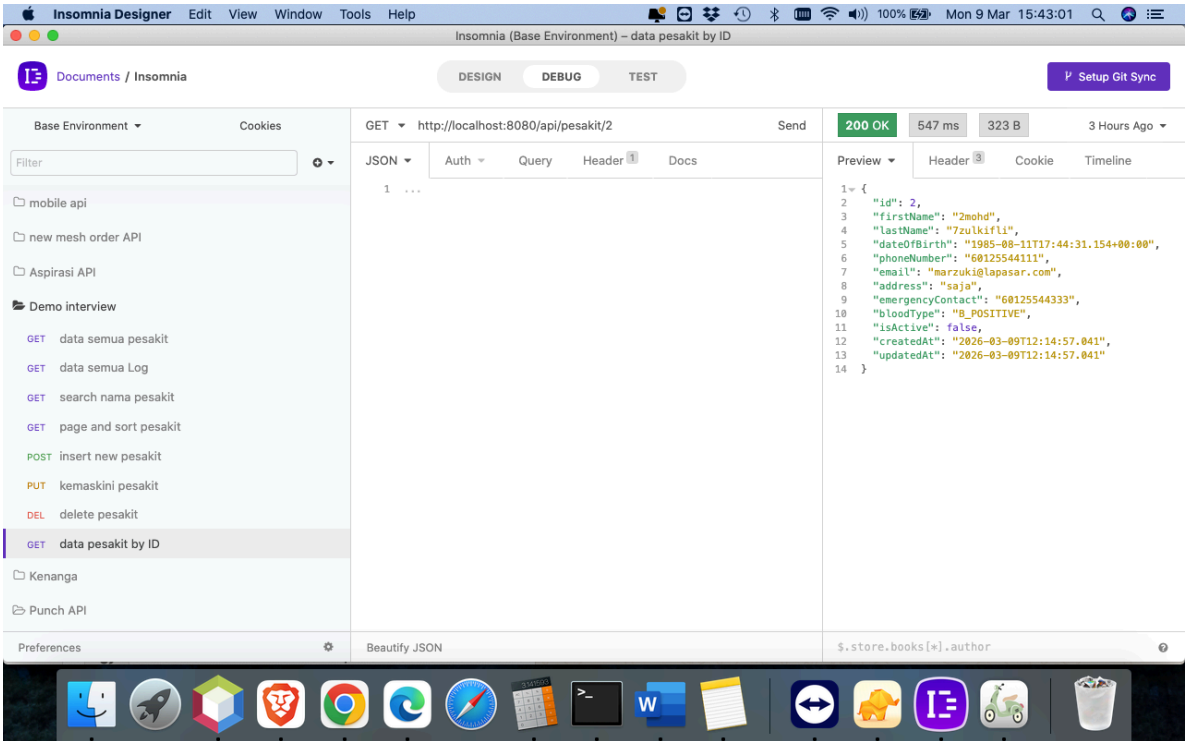
Search name patient



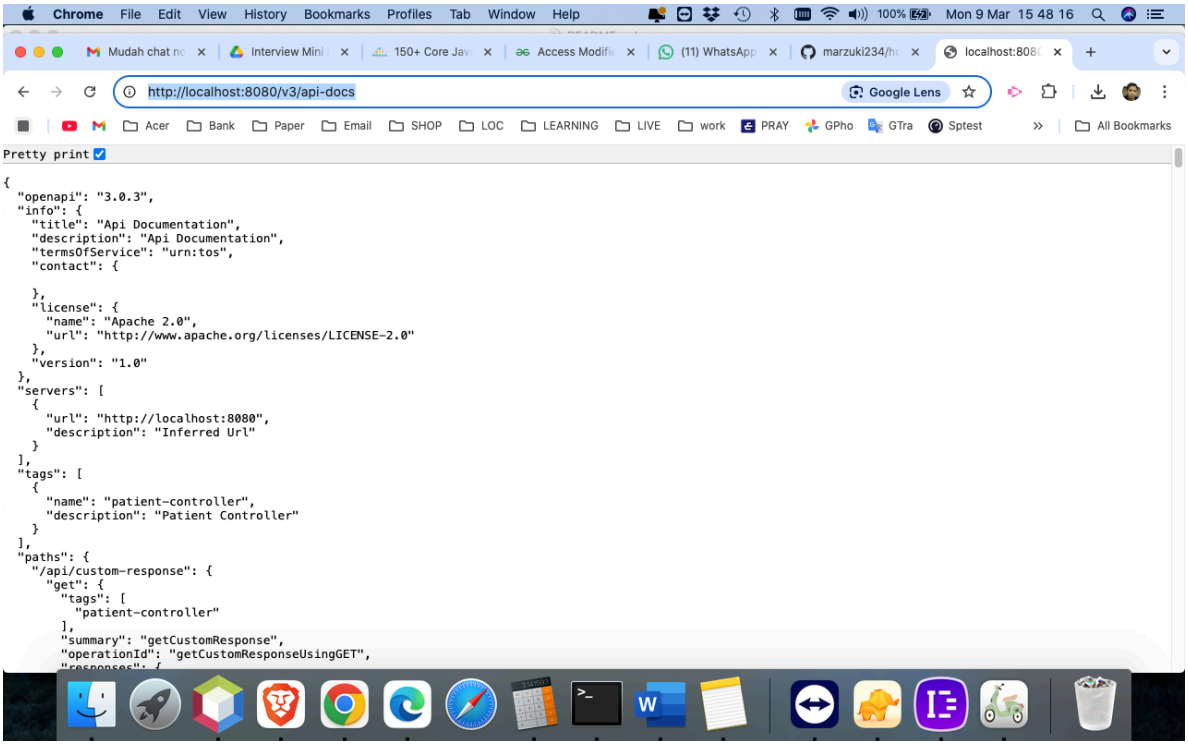
Page and sort list all patients



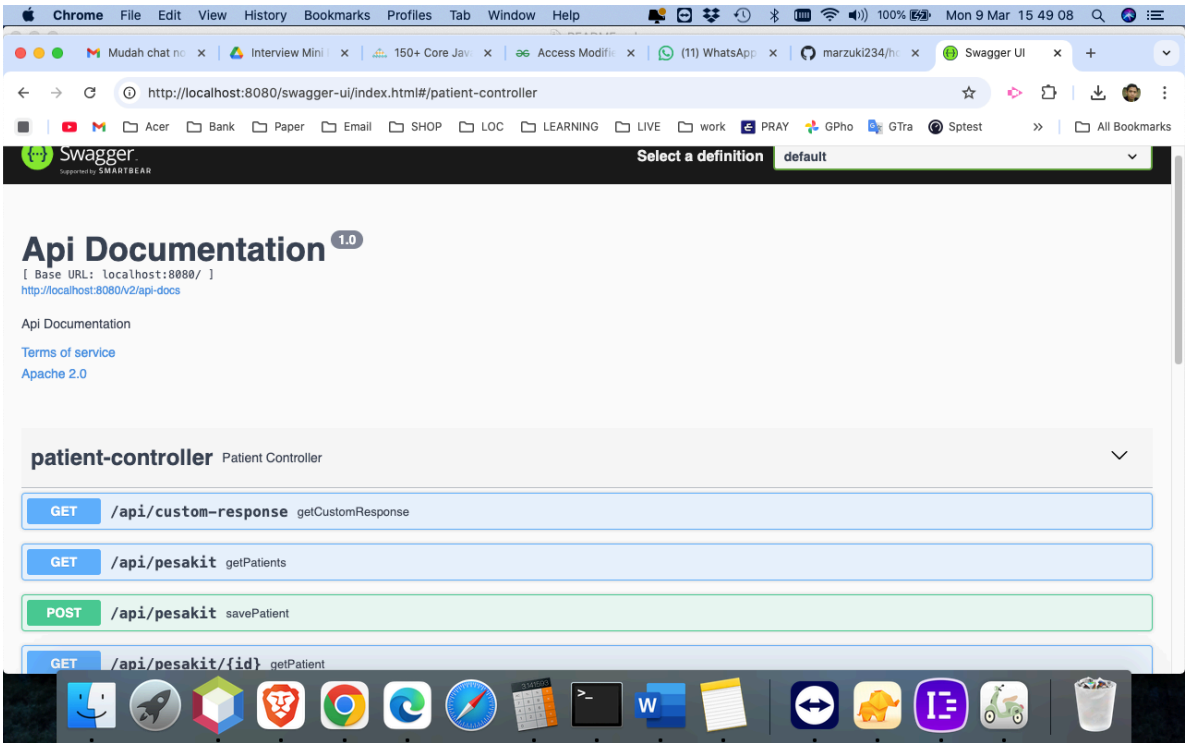
Soft delete patient



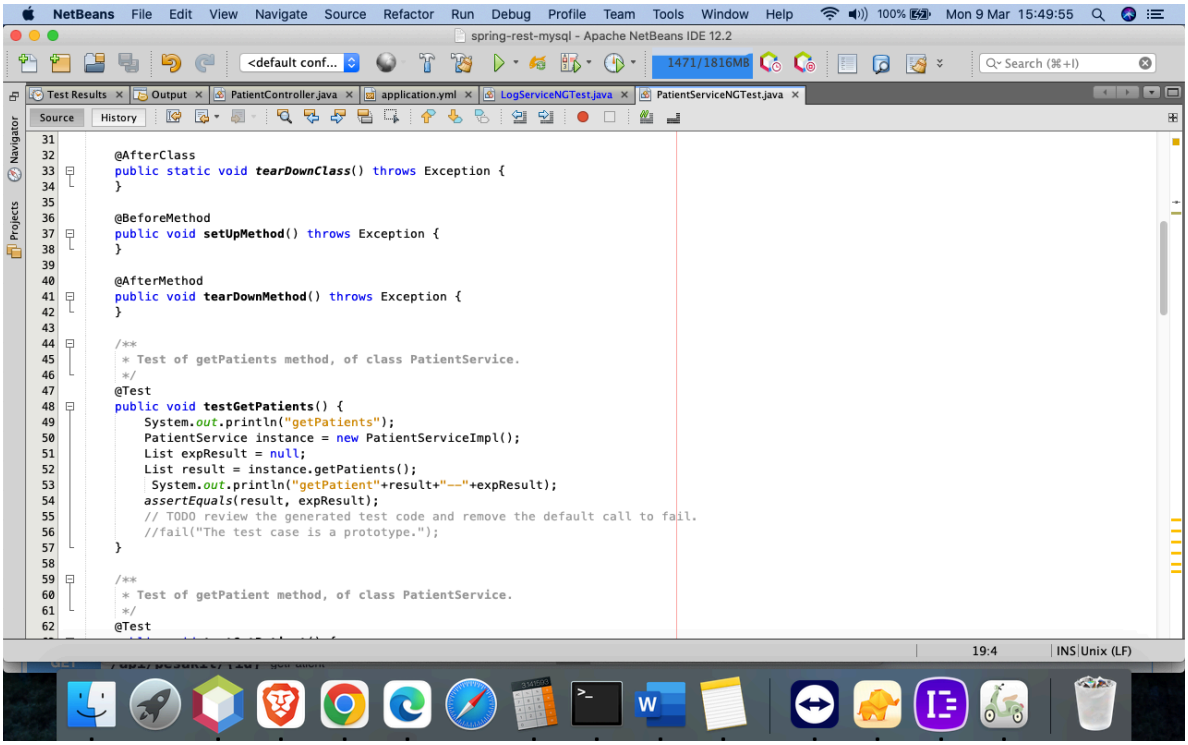
Patient record by id



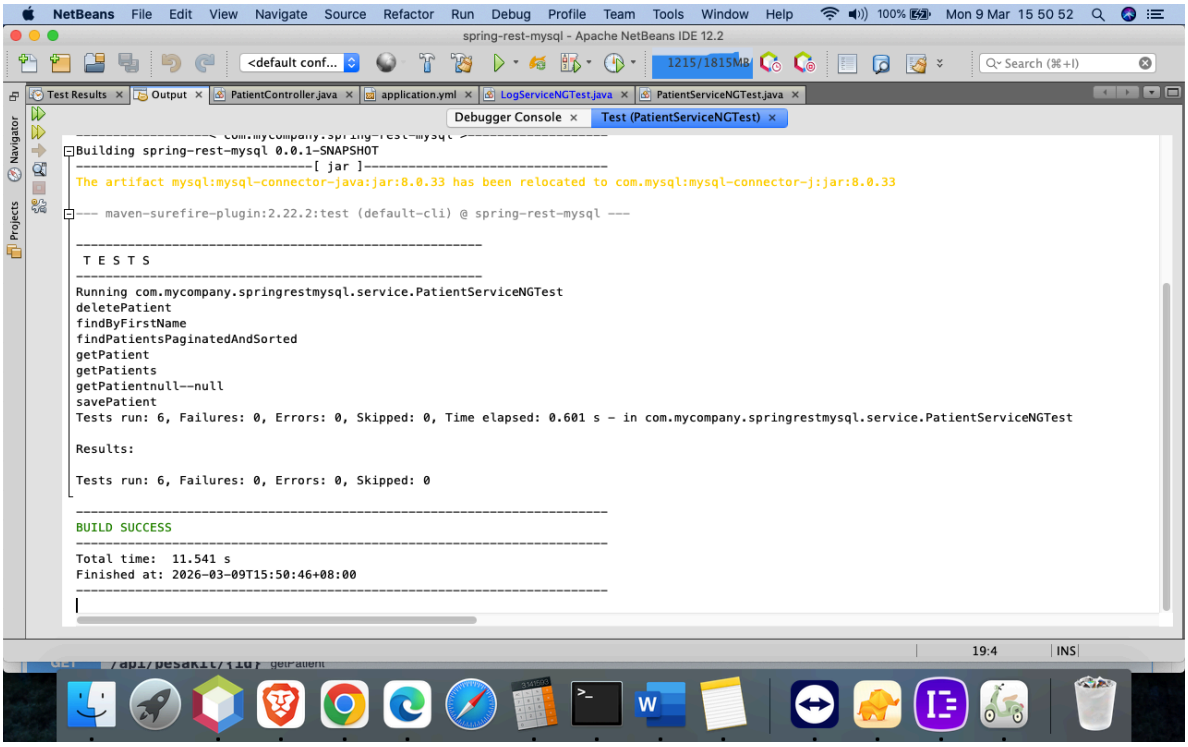

```
openapi": "3.0.3
```



Swagger API documentation



Unit test(integration) patient service



Result Unit test(integration) patient service

Spring Boot RESTful CRUD API Example with MySQL Database

Steps to ON server spring boot 2.5:

1.Go to your project folder run:
 mvn clean package -DskipTests

2.Go to your project folder run:

 mvn clean spring-boot:run

now apache is up.can access

List API endpoint url:
1.http://localhost:8080/api/pesakit(lists all patients)
Method:GET
Output format:JSON
input:no input
output:[

  {
    "id": 1,
    "firstName": "Ahmad",
    "lastName": "Marzuki",
    "dateOfBirth": "1985-11-07T16:00:00.000+00:00",
    "phoneNumber": "60125544123",
    "email": "marzuki@lapasar.com",
    "address": "no 16,persiaran surian..",
    "emergencyContact": "60125544123",
    "bloodType": "AB_NEGATIVE",
    "isActive": true,
    "createdAt": null,
    "updatedAt": null
  }

]

2.http://localhost:8080/log/log(lists all logs audit trail)
Method:GET
Output format:JSON
input:no input
output:[

  {
    "id": 1,
    "firstName": "SAVENEWPATIENT",
    "lastName":
"CreatePatient(firstName=e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49599
1b7852b855mohd, lastName=2zulkifli, dateOfBirth=Mon Aug 12 01:44:31 MYT 1985,
phoneNumber=60125544111, email=marzuki@lapasar.com, address=saja,
emergencyContact=60125544333, bloodType=null, isActive=true)",
    "createdAt": "2026-03-08T00:53:58.789"
  }

]
3.http://localhost:8080/api/pesakit/by-name?firstName=ahmad (search Nama pesakit)
Method:GET
Output format:JSON
input:no input
output:[

  {
    "id": 1,
    "firstName": "SAVENEWPATIENT",
    "lastName":
"CreatePatient(firstName=e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49599
1b7852b855mohd, lastName=2zulkifli, dateOfBirth=Mon Aug 12 01:44:31 MYT 1985,
phoneNumber=60125544111, email=marzuki@lapasar.com, address=saja,
emergencyContact=60125544333, bloodType=null, isActive=true)",
    "createdAt": "2026-03-08T00:53:58.789"
  }

]
4.http://localhost:8080/api/pesakit/pageAndSortPesakit?page=0&size=5&sortBy=email&so
rtDirection=desc (lists all patients ada paging dan bleh sort)
Method:GET
Output format:JSON
input:no input
output:

{
  "content": [
    {
      "id": 1,
      "firstName": "Ahmad",
      "lastName": "Marzuki",
      "dateOfBirth": "1985-11-07T16:00:00.000+00:00",
      "phoneNumber": "60125544123",
      "email": "marzuki@lapasar.com",
      "address": "no 16,persiaran surian..",
      "emergencyContact": "60125544123",
      "bloodType": "AB_NEGATIVE",
      "isActive": true,
      "createdAt": null,
      "updatedAt": null
    },
    {
      "id": 2,
      "firstName": "2mohd",
      "lastName": "2zulkifli",
      "dateOfBirth": "1985-08-11T17:44:31.154+00:00",
      "phoneNumber": "60125544111",
      "email": "marzuki@lapasar.com",
      "address": "saja",
      "emergencyContact": "60125544333",
      "bloodType": null,
      "isActive": true,

```
    "createdAt": "2026-03-07T14:36:35.39",
    "updatedAt": null
  },
  {
    "id": 3,
    "firstName": "3mohd",
    "lastName": "3zulkifli",
    "dateOfBirth": "1985-08-11T17:44:31.154+00:00",
    "phoneNumber": "60125544111",
    "email": "marzuki@lapasar.com",
    "address": "saja",
    "emergencyContact": "60125544333",
    "bloodType": null,
    "isActive": true,
    "createdAt": "2026-03-07T14:36:35.39",
    "updatedAt": null
  },
  {
    "id": 4,
    "firstName": "4mohd",
    "lastName": "4zulkifli",
    "dateOfBirth": "1985-08-11T17:44:31.154+00:00",
    "phoneNumber": "60125544111",
    "email": "marzuki@lapasar.com",
    "address": "saja",
    "emergencyContact": "60125544333",
    "bloodType": null,
    "isActive": true,
    "createdAt": "2026-03-07T14:36:35.39",
    "updatedAt": null
  },
  {
    "id": 5,
    "firstName": "5mohd",
    "lastName": "5zulkifli",
    "dateOfBirth": "1985-08-11T17:44:31.154+00:00",
    "phoneNumber": "60125544111",
    "email": "marzuki@lapasar.com",
    "address": "saja",
    "emergencyContact": "60125544333",
    "bloodType": null,
    "isActive": true,
    "createdAt": "2026-03-07T14:36:35.39",
    "updatedAt": null
  }
],
"pageable": {
  "sort": {
    "empty": false,
    "unsorted": false,
```

"sorted": true
},
"offset": 0,
"pageNumber": 0,
"pageSize": 5,
"unpaged": false,
"paged": true
},
"totalPages": 2,
"totalElements": 7,
"last": false,
"numberOfElements": 5,
"size": 5,
"number": 0,
"sort": {
"empty": false,
"unsorted": false,
"sorted": true
},
"first": true,
"empty": false
}

5.http://localhost:8080/api/pesakit (insert new patients)
Method:POST
Output format:JSON

input:

{
"firstName": "2mohd",
"lastName": "7zulkifli",
"dateOfBirth": "1985-08-11T17:44:31.154",
"phoneNumber": "60125544111",
"email": "marzuki@lapasar.com",
"address": "saja",
"emergencyContact": "60125544333",
"bloodType": "B_POSITIVE",
"isActive": true
}
Output:
{
"id": 2,
"firstName":
"e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855mohd",
"lastName": "2zulkifli",
"dateOfBirth": "1985-08-11T17:44:31.154+00:00",
"phoneNumber": "60125544111",
"email": "marzuki@lapasar.com",
"address": "saja",

"emergencyContact": "60125544333",
"bloodType": null,
"isActive": true,
"createdAt": "2026-03-08T00:53:58.789",
"updatedAt": null
}
6.http://localhost:8080/api/pesakits/1 (kemaskini patients)
Method:PUT
Output format:JSON

input:

{
"id": 1,
"firstName": "vb",
"lastName": "B",
"phoneNumber": "0125544111",
"email": "a@a.com",
"address": "saja",
"emergencyContact": "ali 0123322212",
"isActive": true,
"dateOfBirth": "1985-08-11T17:44:31.154",
"bloodType": "A_POSITIVE"

}
Output:
{
"id": 1,
"firstName": "aa",
"lastName": "B",
"dateOfBirth": "1985-08-11T17:44:31.154+00:00",
"phoneNumber": "0125544111",
"email": "a@a.com",
"address": "saja",
"emergencyContact": "ali 0123322212",
"bloodType": null,
"isActive": true,
"createdAt": null,
"updatedAt": "2026-03-07T17:49:47.602"
}

7.http://localhost:8080/api/pesakits/1 (soft delete patients)
Method:DELETE
Output format:JSON

input:

{
"id": 2,
"isActive": false

}

Output:

{
"id": 2,
"firstName": "2mohd",
"lastName": "2zulkifli",
"dateOfBirth": "1985-08-11T17:44:31.154+00:00",
"phoneNumber": "60125544111",
"email": "marzuki@lapasar.com",
"address": "saja",
"emergencyContact": "60125544333",
"bloodType": null,
"isActive": false,
"createdAt": "2026-03-07T13:19:29.371",
"updatedAt": "2026-03-07T13:19:29.371"
}

8.http://localhost:8080/api/pesakit/1 (data patients by ID)
Method:GET
Output format:JSON

Input:no input

Output:

{
"id": 1,
"firstName": "Ahmad",
"lastName": "Marzuki",
"dateOfBirth": "1985-11-07T16:00:00.000+00:00",
"phoneNumber": "60125544123",
"email": "marzuki@lapasar.com",
"address": "no 16,persiaran surian..",
"emergencyContact": "60125544123",
"bloodType": "AB_NEGATIVE",
"isActive": true,
"createdAt": null,
"updatedAt": null
}

openapi 3 url:

http://localhost:8080/v3/api-docs

http://localhost:8080/swagger-ui/index.html

https://www.baeldung.com/spring-rest-openapi-documentation

integration test(JUNIT4 test)
/springboot-rest-
mysql/src/test/java/com/mycompany/springrestmysql/service/PatientServiceIT.java